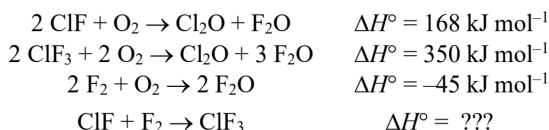


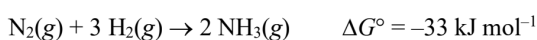
Thermodynamics

1. What is the standard enthalpy change for the reaction of ClF with F₂ to form ClF₃? (All species are in the gas phase.)



- (A) -46 kJ mol^{-1} (B) -69 kJ mol^{-1}
(C) -114 kJ mol^{-1} (D) -136 kJ mol^{-1}

2. At 25 °C, the standard free energy change for the gas-phase reaction of hydrogen and nitrogen to form NH₃ is $-33.0 \text{ kJ mol}^{-1}$.



Under what circumstances is $\Delta G > 0$ at 298 K?

- (A) Partial pressures of N₂ and H₂ = 0.1 bar, partial pressure of NH₃ = 10 bar
(B) Partial pressures of N₂ and H₂ = 10 bar, partial pressure of NH₃ = 0.1 bar
(C) All partial pressures = 1×10^{-2} bar
(D) Under no possible circumstances

3. In a coal-fired power plant, coal is burned to generate heat that boils water. 2.5 kg of coal is burned to heat 500.0 kg of water at 95 °C, ultimately converting 30.0 kg to steam. Which type of coal was burned? (For liquid water near 100 °C, $C_p = 4.21 \text{ kJ kg}^{-1} \text{ K}^{-1}$ and $\Delta H^\circ_{\text{vap}} = 2260 \text{ kJ kg}^{-1}$.)

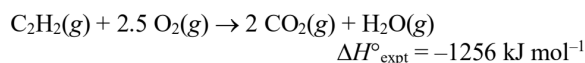
- (A) Lignite, $\Delta H^\circ_{\text{combustion}} = 15 - 20 \text{ kJ g}^{-1}$
(B) Sub-bituminous, $\Delta H^\circ_{\text{combustion}} = 23 - 28 \text{ kJ g}^{-1}$
(C) Bituminous, $\Delta H^\circ_{\text{combustion}} = 26 - 30 \text{ kJ g}^{-1}$
(D) Anthracite, $\Delta H^\circ_{\text{combustion}} = 30 - 33 \text{ kJ g}^{-1}$

4. In a vacuum calorimetry experiment, high-energy plasma discharges between Ti(s) electrodes are used to electrolyze water in dilute KOH(aq). During an experiment, the electrical input is 500. kJ and the calorimeter records 610. kJ of heat release into the surroundings. Which best explains this experimental result?

- (A) The electrodes undergo partial oxidation, an additional reaction not accounted for in electrolysis.
(B) The system's large specific heat capacity increases the total heat registered per unit of electrical energy.
(C) The overall process is endothermic, so the surroundings must release heat to maintain equilibrium.
(D) The entropy increase from gas formation causes the system to do work, resulting in additional temperature rise.

5. The standard enthalpy of combustion of ethyne can be estimated from the tabulated bond dissociation enthalpies (BDE) shown below to be -982 kJ mol^{-1} . The experimental $\Delta H^\circ_{\text{combustion}}$ is $-1256 \text{ kJ mol}^{-1}$. Which is the best explanation for the discrepancy?

Bond	BDE (kJ mol ⁻¹)
C≡C	839
C–H	412
O=O	498
C=O	741
O–H	463



- (A) The tabulated C–H BDE does not account for the unusual strength of the bond formed to the *sp*-hybridized carbon in C₂H₂.
(B) The tabulated O=O BDE does not account for the unusual weakness of the bond in dioxygen due to its paramagnetism.
(C) The tabulated C=O BDE does not account for the stabilization due to electronic delocalization in carbon dioxide.
(D) The tabulated O–H BDE does not account for the stabilization due to hydrogen bonding in water.

6. For which reaction is $\Delta S^\circ > 0$?

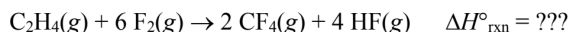
- (A) $\text{P}_4(\text{s}) \rightarrow 4 \text{ P}(\text{s, red})$
(B) $\text{CO}_2(\text{s}) \rightarrow \text{CO}_2(\text{g})$
(C) $3 \text{ O}_2(\text{g}) \rightarrow 2 \text{ O}_3(\text{g})$
(D) $2 \text{ SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{ SO}_3(\text{g})$

Thermodynamics

7. The standard enthalpy of fusion of 1,4-dichlorobenzene is 18.2 kJ mol^{-1} and its standard entropy of fusion is $55.3 \text{ J mol}^{-1} \text{ K}^{-1}$. What is the melting point of 1,4-dichlorobenzene?

(A) 37.1°C (B) 56.0°C
(C) 101°C (D) 329°C

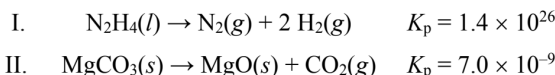
8. What is $\Delta H^\circ_{\text{rxn}}$ for the gas-phase reaction of ethene and fluorine shown below?



Substance	ΔH°_f , kJ mol^{-1}
$\text{C}_2\text{H}_4(\text{g})$	52.4
$\text{CF}_4(\text{g})$	-679.0
$\text{HF}(\text{g})$	-273.3

(A) $-2503.6 \text{ kJ mol}^{-1}$ (B) $-1004.7 \text{ kJ mol}^{-1}$
(C) $1004.7 \text{ kJ mol}^{-1}$ (D) $2503.6 \text{ kJ mol}^{-1}$

9. Which reactions are spontaneous under standard conditions?



(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

10. Solid gallium melts at 29.8°C with an enthalpy of fusion of 5.59 kJ mol^{-1} . The heat capacity of solid gallium is $26.1 \text{ J mol}^{-1} \text{ K}^{-1}$ and the heat capacity of liquid gallium is $28.5 \text{ J mol}^{-1} \text{ K}^{-1}$. If equal masses of solid gallium at 20.0°C and liquid gallium at 60.0°C are mixed in a well-insulated container, what is the temperature after the mixture has attained equilibrium?

(A) 29.8°C (B) 40.0°C
(C) 40.4°C (D) 40.9°C

11. What is the K_{sp} of AgBr at 298 K ?

Species	ΔG°_f (298 K), kJ mol^{-1}
$\text{Ag}^+(\text{aq})$	77.1
$\text{Br}^-(\text{aq})$	-102.8
$\text{AgBr}(\text{s})$	-96.9

(A) 3.3×10^{-13} (B) 3.1×10^{-14}
(C) 1.0×10^{-17} (D) 3.2×10^{-22}

12. For a chemical reaction under conditions where it is at equilibrium, which statements must be true?

I. $\Delta S_{\text{rxn}} > 0$ II. $\Delta G_{\text{rxn}} = 0$

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

13. Which statement about phase transitions is correct?

(A) Melting is endothermic and freezing is exothermic.
(B) Vaporization is exothermic and condensation is endothermic.
(C) The enthalpy change for a spontaneous phase transition is always positive.
(D) The entropy change for a spontaneous phase transition is always positive.

14. Based on the given bond dissociation enthalpies (BDEs), what is $\Delta H^\circ_{\text{rxn}}$ for the addition of methane to ethyne to give propene?



Bond	BDE, kJ mol^{-1}	Bond	BDE, kJ mol^{-1}
C-H	415	C=C	611
C-C	345	C≡C	837

(A) -119 kJ mol^{-1} (B) -196 kJ mol^{-1}
(C) -259 kJ mol^{-1} (D) -296 kJ mol^{-1}

15. Which statements must be true for a nonspontaneous gas-phase chemical reaction at constant temperature and pressure?

I. $\Delta G > 0$ II. $K_p < 1$

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

16. Benzene, C_6H_6 , has $\Delta H^\circ_{\text{vap}} = 30.7 \text{ kJ mol}^{-1}$ at its normal boiling point of 80.1°C . What is $\Delta S^\circ_{\text{vap}}$ at this temperature?

(A) $10.5 \text{ J mol}^{-1} \text{ K}^{-1}$ (B) $86.9 \text{ J mol}^{-1} \text{ K}^{-1}$
(C) $255 \text{ J mol}^{-1} \text{ K}^{-1}$ (D) $383 \text{ J mol}^{-1} \text{ K}^{-1}$

17. Which reaction has $\Delta S^\circ < 0$?

(A) $\text{C}(\text{s, graphite}) \rightarrow \text{C}(\text{s, diamond})$
(B) $\text{N}_2\text{O}_4(\text{g}) \rightarrow 2 \text{NO}_2(\text{g})$
(C) $\text{I}_2(\text{s}) \rightarrow \text{I}_2(\text{g})$
(D) $\text{NH}_4^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq}) \rightarrow \text{NH}_3(\text{aq}) + \text{CH}_3\text{COOH}(\text{aq})$

Thermodynamics

18. What is the K_{sp} of $PbCl_2$ at 298 K?

Species	ΔH°_f , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
$Pb^{2+}(aq)$	-1.7	10.5
$PbCl_2(s)$	-359.4	136.0
$Cl^-(aq)$	-167.2	56.5

- (A) 1.2×10^{-11} (B) 1.8×10^{-5}
(C) 8.2×10^{-5} (D) 3.7×10^{-4}

19. Given the standard enthalpies of the gas-phase reactions of hydrogen sulfide, what is the standard enthalpy of reaction for the gas-phase combustion of carbon disulfide?

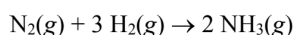
Reaction	ΔH°_{rxn} , kJ mol ⁻¹
$H_2S + 1.5 O_2 \rightarrow SO_2 + H_2O$	-518.2
$2 H_2S + CO_2 \rightarrow CS_2 + 2 H_2O$	67.8
$CS_2 + 3 O_2 \rightarrow CO_2 + 2 SO_2$	???

- (A) -450.4 kJ mol⁻¹ (B) -586.0 kJ mol⁻¹
(C) -968.6 kJ mol⁻¹ (D) -1104.2 kJ mol⁻¹

20. Which substance has a standard enthalpy of formation, ΔH°_f , that is greater than zero?

- (A) $O_2(g)$ (B) $F_2(g)$ (C) $S_2(g)$ (D) $Cl_2(g)$

21. For the reaction



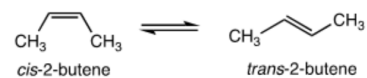
$\Delta G_{rxn} = -33$ kJ mol⁻¹ when all gases have a partial pressure of 1.0 bar. Changing the pressure of which gas to 2.0 bar will give the most negative value of ΔG_{rxn} ?

- (A) N_2
(B) H_2
(C) NH_3
(D) The value of ΔG_{rxn} will be the same regardless of the gas pressures.

22. Gallium metal melts at 29.8 °C with $\Delta H^\circ_{fusion} = 5.59$ kJ mol⁻¹. If equal masses of solid gallium initially at 20.0 °C and liquid gallium initially at 35.0 °C are mixed in a well-insulated container, what percentage of the gallium will be solid after the mixture has reached equilibrium? The molar heat capacity of solid gallium is 26.2 J mol⁻¹ K⁻¹ and the molar heat capacity of liquid gallium is 28.0 J mol⁻¹ K⁻¹.

- (A) 34.7% (B) 49.0% (C) 51.0% (D) 65.3%

23. *cis*-2-butene isomerizes to *trans*-2-butene with $\Delta G^\circ_{rxn} = -3.0$ kJ mol⁻¹ at 298 K.

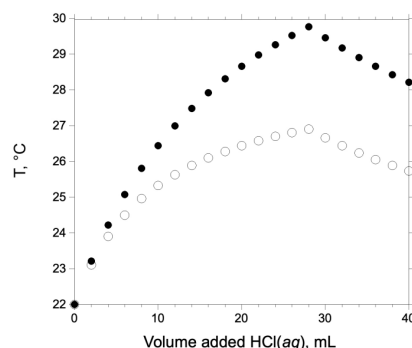


$$\Delta G^\circ_{rxn} = -3.0 \text{ kJ mol}^{-1}$$

Some *cis*-2-butene is introduced into a vessel maintained at 298 K, along with a catalyst for isomerization. After equilibrium is attained, what is the mole fraction of *cis*-2-butene in the vessel?

- (A) 0.00 (B) 0.23 (C) 0.30 (D) 0.50

24. A certain volume of $NaOH(aq)$ is titrated against $HCl(aq)$ in a well-insulated vessel with constant stirring. Each solution is initially at 22.0 °C, and the temperature of the reaction mixture is monitored after each 2.0 mL of hydrochloric acid is added.



The data in closed circles correspond to an experiment conducted with 20.0 mL $NaOH$ solution and 1.0 M HCl . The open circles were carried out using the same concentration but a different volume of $NaOH$ solution, and using a different concentration of aqueous HCl . Which volume of $NaOH$ and concentration of HCl correspond to the conditions that give the open circles?

	Volume $NaOH$	Concentration HCl
Closed circles:	20 mL	1.0 M
Open circles:		
(A)	40 mL	2.0 M
(B)	30 mL	2.0 M
(C)	15 mL	0.5 M
(D)	10 mL	0.5 M

25. The amount of energy released by a chemical process is always equal to that absorbed by the system and surroundings. This is a consequence of which law?

- (A) The Zeroth Law of Thermodynamics
(B) The First Law of Thermodynamics
(C) The Second Law of Thermodynamics
(D) The Third Law of Thermodynamics

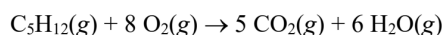
Thermodynamics

26. What is ΔG°_f of NO(g) at 298 K?

Substance	ΔH°_f , kJ mol^{-1}	S° , $\text{J mol}^{-1} \text{K}^{-1}$
$\text{N}_2(\text{g})$	0	191.6
$\text{O}_2(\text{g})$	0	205.2
NO(g)	90.3	210.8

- (A) 27.5 kJ mol^{-1} (B) 86.6 kJ mol^{-1}
(C) $145.7 \text{ kJ mol}^{-1}$ (D) $153.1 \text{ kJ mol}^{-1}$

27. Which statements about entropy, enthalpy, and energy are correct for the reaction below?



- I. $\Delta S^\circ > 0$ II. $\Delta H^\circ > \Delta E^\circ$

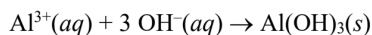
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

28. 4.00 mL of 2.00 M aqueous NaOH is added to 100.0 mL of 0.200 M aqueous NaHSO_4 , all initially at 25.00°C , in a well-insulated flask. Neglecting the effects of dilution, and assuming that volumes are additive and that all solutions have the same density and specific heat capacity as pure water, what is the final temperature of the mixture?

Species	ΔH°_f , kJ mol^{-1}
$\text{OH}^-(\text{aq})$	-230
$\text{HSO}_4^-(\text{aq})$	-886
$\text{SO}_4^{2-}(\text{aq})$	-909
$\text{H}_2\text{O}(\text{l})$	-286

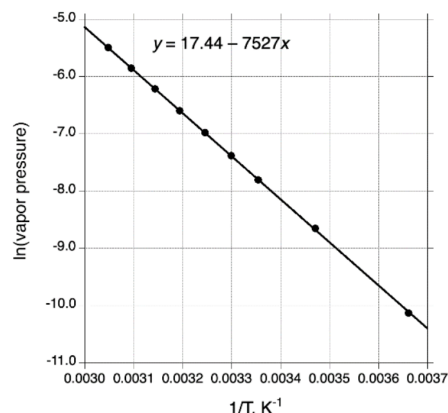
- (A) 25.18°C (B) 26.06°C
(C) 26.45°C (D) 28.63°C

29. The K_{sp} of $\text{Al}(\text{OH})_3$ is 2.0×10^{-31} at 298 K. What is ΔG° (at 298 K) for the precipitation of $\text{Al}(\text{OH})_3$ according to the equation below?



- (A) -175 kJ mol^{-1} (B) 14.7 kJ mol^{-1}
(C) 70.6 kJ mol^{-1} (D) 175 kJ mol^{-1}

30. The vapor pressure of solid I_2 is measured (in bar) over the temperature range 0 – 55°C . The natural logarithms of these vapor pressures are plotted as a function of the reciprocal of the absolute temperature as shown below. What is the standard enthalpy of sublimation ($\Delta H^\circ_{\text{subl}}$) of $\text{I}_2(\text{s})$?

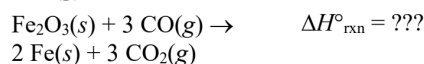
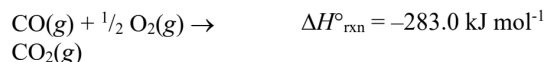


- (A) 7.53 kJ mol^{-1} (B) 17.4 kJ mol^{-1}
(C) 19.4 kJ mol^{-1} (D) 62.6 kJ mol^{-1}

31. A container with 100.0 g of ice at 0°C is placed in a humid room whose temperature is 40°C . The ice melts as water vapor condenses into the container. Assuming that all the heat transferred to the container comes from condensation, how much water will have condensed in the container once all the ice is melted and has reached 40°C ? The heat of fusion of ice is 334 J g^{-1} and the heat of vaporization of water is 2260 J g^{-1} .

- (A) 13.8 g (B) 14.8 g (C) 22.2 g (D) 677 g

32. The standard enthalpy of formation of hematite, $\text{Fe}_2\text{O}_3(\text{s})$, is $-825.5 \text{ kJ mol}^{-1}$. What is the standard enthalpy of reaction for the reduction of hematite with carbon monoxide to give metallic iron as shown below?



- (A) $-23.5 \text{ kJ mol}^{-1}$ (B) $-542.5 \text{ kJ mol}^{-1}$
(C) $-849.0 \text{ kJ mol}^{-1}$ (D) $-1674.5 \text{ kJ mol}^{-1}$

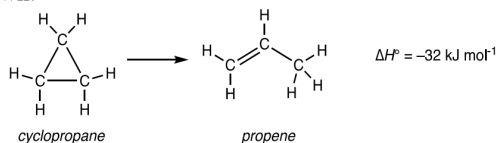
33. Which of the following equations correctly describe the change in internal energy, ΔE , of a reaction taking place in a bomb calorimeter?

- I. $\Delta E = q$
II. $\Delta E = \Delta H - V\Delta P$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

Thermodynamics

34. Cyclopropane isomerizes exothermically to propene as shown.

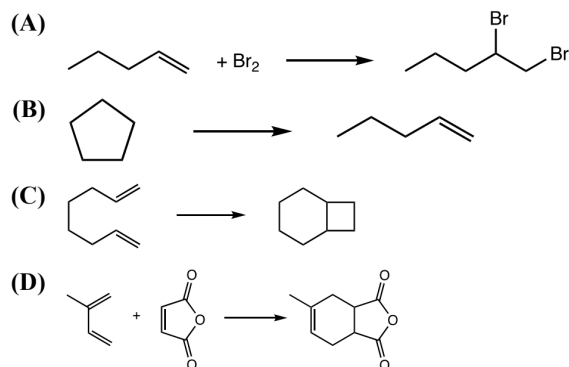


The C–C single bond in propene has an estimated bond dissociation enthalpy (BDE) of 345 kJ mol⁻¹, while the C=C double bond in propene has a BDE of 611 kJ mol⁻¹. Assuming that the C–H bonds in propene and cyclopropane all have the same BDE, what is the estimated BDE of a C–C single bond in cyclopropane?

- (A) 308 kJ mol⁻¹ (B) 329 kJ mol⁻¹
(C) 423 kJ mol⁻¹ (D) 924 kJ mol⁻¹
35. Two reactions have similar $\Delta H^\circ_{\text{rxn}}$ values ($\Delta H^\circ_{\text{rxn}(1)} \approx \Delta H^\circ_{\text{rxn}(2)}$), but reaction (1) has a much smaller standard entropy change than reaction (2) ($\Delta S^\circ_{\text{rxn}(1)} \ll \Delta S^\circ_{\text{rxn}(2)}$). At 298 K, which statements about these two reactions must be correct?
- Reaction (1) must have a larger equilibrium constant ($K_{\text{eq}(1)} > K_{\text{eq}(2)}$).
 - Reaction (1) must have a larger Arrhenius prefactor ($A_{(1)} > A_{(2)}$).

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

36. Which reaction has $\Delta S^\circ_{\text{rxn}} > 0$? All reactants and products are liquids.



37. Burning 48.0 g of graphite in excess oxygen under standard conditions releases 1574.0 kJ of heat. What is ΔH°_f of CO₂(g)?

- (A) 1574 kJ mol⁻¹ (B) -1574 kJ mol⁻¹
(C) -394 kJ mol⁻¹ (D) 32.8 kJ mol⁻¹

38. Two metal samples, labeled A and B, absorb the same amount of heat. Sample A has a mass of 10.0 g, and its temperature increases by 38 °C. Sample B has a mass of 20.0 g, and its temperature increases by 23 °C. Which sample has the greater specific heat capacity?

- (A) Sample A
(B) Sample B
(C) Both samples have the same specific heat capacity.
(D) It is impossible to determine from the information given.

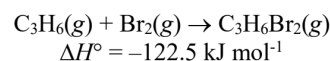
39. A reaction has $\Delta S^\circ > 0$ and $\Delta H^\circ > 0$. Which statement about this reaction must be correct?

- (A) If carried out in a well-insulated flask, the temperature of the reaction mixture will decrease.
(B) It will occur spontaneously at 298 K and 1 atm pressure.
(C) As the temperature is raised, K_{eq} for this reaction decreases.
(D) The reaction has more moles of products than it has moles of reactants.

40. What is the boiling point of water in a pressure cooker with a pressure of 2.00 atm? (The enthalpy of vaporization of water is 40.7 kJ mol⁻¹.)

- (A) 101 °C (B) 121 °C (C) 141 °C (D) 200 °C

41. The gas-phase bromination of propene has a standard enthalpy of reaction of -122.5 kJ mol⁻¹.



Substance	ΔH°_f , kJ mol ⁻¹
C ₃ H ₆ (g)	20.4
Br ₂ (g)	30.9

What is the standard enthalpy of formation of C₃H₅Br₂(g)?

- (A) -71.2 kJ mol⁻¹ (B) -102.1 kJ mol⁻¹
(C) -142.9 kJ mol⁻¹ (D) -173.8 kJ mol⁻¹

42. Titanium has a normal melting point of 1668 °C and a molar enthalpy of fusion of 14.15 kJ mol⁻¹. The standard molar entropy of liquid titanium is 97.53 J mol⁻¹ K⁻¹ at 1668 °C. What is the standard molar entropy of solid titanium at this temperature?

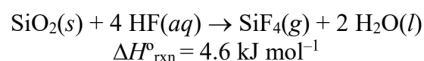
- (A) 89.05 J mol⁻¹ K⁻¹ (B) 90.24 J mol⁻¹ K⁻¹
(C) 97.52 J mol⁻¹ K⁻¹ (D) 104.82 J mol⁻¹ K⁻¹

Thermodynamics

43. For which reaction is $\Delta H^\circ_{\text{rxn}}$ equal to ΔH°_f of $\text{CaSO}_3(s)$?
- (A) $\text{Ca}(s) + \frac{1}{8} \text{S}_8(s) + \frac{3}{2} \text{O}_2(g) \rightarrow \text{CaSO}_3(s)$
 (B) $8 \text{Ca}(s) + \text{S}_8(s) + 12 \text{O}_2(g) \rightarrow 8 \text{CaSO}_3(s)$
 (C) $8 \text{Ca}(s) + \text{S}_8(g) + 12 \text{O}_2(g) \rightarrow 8 \text{CaSO}_3(s)$
 (D) $\text{CaO}(s) + \text{SO}_2(g) \rightarrow \text{CaSO}_3(s)$
44. A 2.00 g sample of ice at 0.0 °C is placed in 50.0 g of water initially at 25.0 °C in an insulated container. What is the final temperature after the system has achieved equilibrium? (For ice, $\Delta H^\circ_{\text{fusion}} = 6.01 \text{ kJ mol}^{-1}$.)
- (A) 21.0 °C (B) 21.8 °C
 (C) 22.5 °C (D) 24.0 °C
45. Ethanol has a normal boiling point of 78.3 °C and a standard heat of vaporization ($\Delta H^\circ_{\text{vap}}$) of 38.6 kJ mol⁻¹. What is the vapor pressure of ethanol at 45.0 °C?
- (A) 0.25 atm (B) 0.57 atm
 (C) 0.87 atm (D) 0.91 atm
46. If the average carbon-hydrogen bond dissociation enthalpy in ethane is 416 kJ mol⁻¹, what is the bond dissociation enthalpy of the carbon-carbon bond in ethane?
- | Species | ΔH°_f , kJ mol ⁻¹ |
|---------------------------|---|
| $\text{C}_2\text{H}_6(g)$ | -84.7 |
| $\text{H}(g)$ | 217.9 |
| $\text{C}(g)$ | 718.4 |
- (A) 164 kJ mol⁻¹ (B) 333 kJ mol⁻¹
 (C) 386 kJ mol⁻¹ (D) 404 kJ mol⁻¹
47. A reaction has $K_{\text{eq}} = 20$ at 298 K, and K_{eq} increases with increasing temperature between 298 K and 350 K. What may be concluded from these observations?
- I. $\Delta G^\circ_{\text{rxn}} < 0$ at 330 K
 II. $\Delta S^\circ_{\text{rxn}} > 0$ at 330 K
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
48. A sample of water in a movable piston is maintained at 100.0 °C with an external pressure p . Which statement best describes the transformation of some of the liquid water into water vapor?
- (A) It is spontaneous if $p \geq 1 \text{ atm}$.
 (B) It is reversible if $p = 1 \text{ atm}$ and spontaneous if $p > 1 \text{ atm}$.
 (C) It is nonspontaneous if $p \geq 1 \text{ atm}$.
 (D) It is reversible if $p = 1 \text{ atm}$ and nonspontaneous if $p > 1 \text{ atm}$.
49. The specific heat capacity of iron is 0.461 J g⁻¹ K⁻¹ and that of titanium is 0.544 J g⁻¹ K⁻¹. A sample consisting of a mixture of 10.0 g Fe and 10.0 g Ti at 100.0 °C loses 200. J of heat to the environment. What is the final temperature of the sample?
- (A) 89.9 °C (B) 80.1 °C
 (C) 60.2 °C (D) 39.8 °C
50. Which sample has the greatest entropy at a given temperature?
- (A) 2 mol HBr(s)
 (B) 2 mol HBr(l)
 (C) 2 mol HBr(g)
 (D) 1 mol H₂(g) + 1 mol Br₂(l)
51. For an endothermic reaction to be spontaneous under standard conditions at constant pressure at some temperature T , which must be true?
- (A) The entropy change ΔS° must be positive and greater than $\Delta H^\circ/T$.
 (B) The entropy change ΔS° must be positive and less than $\Delta H^\circ/T$.
 (C) The entropy change ΔS° must be negative and greater than $-\Delta H^\circ/T$.
 (D) The entropy change ΔS° must be negative and less than $-\Delta H^\circ/T$.
52. For a constant-pressure process, what is the difference between the internal energy change (ΔU) and the enthalpy change (ΔH)?
- (A) Heat (B) Work
 (C) Entropy (D) Gibbs free energy

Thermodynamics

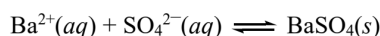
53. The standard enthalpy of reaction for the dissolution of silica in aqueous HF is 4.6 kJ mol^{-1} . What is the standard enthalpy of formation of $\text{SiF}_4(\text{g})$?



Compound	$\text{SiO}_2(\text{s})$	$\text{HF}(\text{aq})$	$\text{H}_2\text{O}(\text{l})$	$\text{SiF}_4(\text{g})$
$\Delta H^\circ_f, \text{ kJ mol}^{-1}$	-910.9	-320.1	-285.8	???

- (A) $-1624.3 \text{ kJ mol}^{-1}$ (B) $-1615.1 \text{ kJ mol}^{-1}$
(C) $-949.8 \text{ kJ mol}^{-1}$ (D) $-940.6 \text{ kJ mol}^{-1}$

54. The K_{sp} of BaSO_4 at 298 K is 1.1×10^{-10} . What is ΔG° at 298 K for the following reaction?



- (A) -57 kJ mol^{-1} (B) -25 kJ mol^{-1}
(C) 25 kJ mol^{-1} (D) 57 kJ mol^{-1}

55. To 100.0 g water at 25.00°C in a well-insulated container is added a block of aluminum initially at 100.0°C . The temperature of the water once the system reaches thermal equilibrium is 28.00°C . What is the mass of the aluminum block? (The specific heat capacity of Al is $0.900 \text{ J g}^{-1} \text{ K}^{-1}$.)

- (A) 4.17 g (B) 18.6 g (C) 19.4 g (D) 130. g

56. The standard enthalpy of formation, ΔH°_f , of the compound $\text{MgO}(\text{s})$ is equal to the standard enthalpy change for which reaction?

- (A) $\text{Mg}(\text{s}) + \frac{1}{2} \text{ O}_2(\text{g}) \rightarrow \text{MgO}(\text{s})$
(B) $2 \text{ Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2 \text{ MgO}(\text{s})$
(C) $\text{Mg}(\text{g}) + \text{O}(\text{g}) \rightarrow \text{MgO}(\text{s})$
(D) $\text{Mg}^{2+}(\text{aq}) + \text{O}^{2-}(\text{aq}) \rightarrow \text{MgO}(\text{s})$

57. The enthalpy of formation of $\text{XeF}_2(\text{g})$ is -108 kJ mol^{-1} and the bond dissociation enthalpy of the F-F bond is 155 kJ mol^{-1} . What is the average bond dissociation enthalpy of a Xe-F bond?

- (A) 47 kJ mol^{-1} (B) 54 kJ mol^{-1}
(C) 132 kJ mol^{-1} (D) 263 kJ mol^{-1}

58. What is the standard Gibbs free energy of formation, ΔG°_f , of $\text{NH}_3(\text{g})$ at 298 K?

Substance	$\Delta H^\circ_f, \text{ kJ mol}^{-1}$	$S^\circ, \text{ J mol}^{-1} \text{ K}^{-1}$
$\text{H}_2(\text{g})$	0	131
$\text{N}_2(\text{g})$	0	192
$\text{NH}_3(\text{g})$	-46	193

- (A) -104 kJ mol^{-1} (B) -16 kJ mol^{-1}
(C) -7 kJ mol^{-1} (D) 13 kJ mol^{-1}

59. The vaporization of a liquid at a certain temperature and pressure is spontaneous. For this process, which of the inequalities regarding the Gibbs free energy G and the internal energy E are correct?

- I. $\Delta G < 0$ II. $\Delta E < \Delta H$
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

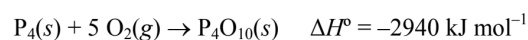
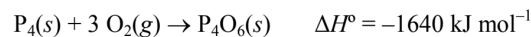
60. Which of the following reactions takes place with an increase in entropy under standard conditions?

- (A) $\text{NH}_4^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq}) \rightarrow \text{NH}_3(\text{aq}) + \text{CH}_3\text{COOH}(\text{aq})$
(B) $\text{CaO}(\text{s}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s})$
(C) $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{Cl}^-(\text{aq})$
(D) $\text{C}_2\text{H}_4(\text{g}) + \text{Br}_2(\text{l}) \rightarrow \text{C}_2\text{H}_4\text{Br}_2(\text{l})$

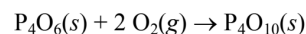
61. A sample of 54.0 g of methanol is heated from 25.0°C to 35.0°C . How much heat is required? The specific heat capacity of methanol is $2.48 \text{ J g}^{-1} \text{ K}^{-1}$.

- (A) 0.00459 J (B) 0.0747 J
(C) 1340 J (D) 4690 J

62. Given the standard enthalpy changes for the reactions:



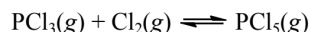
Calculate the standard enthalpy change ΔH° for the following reaction:



- (A) $-4.58 \times 10^3 \text{ kJ mol}^{-1}$ (B) $-1.30 \times 10^3 \text{ kJ mol}^{-1}$
(C) 1.79 kJ mol^{-1} (D) $4.82 \times 10^6 \text{ kJ mol}^{-1}$

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63. At what temperature is the following reaction at equilibrium when all substances are at standard pressure? Assume that entropies and enthalpies of reaction do not vary with temperature.



Substance	ΔH_f° , kJ mol ⁻¹	S° , J mol ⁻¹ K ⁻¹
PCl ₃ (g)	-288.7	311.6
Cl ₂ (g)	0	223.1
PCl ₅ (g)	-374.9	364.2

- (A) 506 K (B) 1640 K
(C) 1980 K (D) 4260 K

64. Given the bond dissociation enthalpies (BDE) below, what is the approximate ΔH_f° for H₂O(g)?

Bond	BDE, kJ mol ⁻¹	Bond	BDE, kJ mol ⁻¹
H-H	432	O-O	146
O-H	467	O=O	495

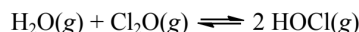
- (A) -934 kJ mol⁻¹ (B) -510. kJ mol⁻¹
(C) -429 kJ mol⁻¹ (D) -255 kJ mol⁻¹

65. Which of the following are true for a spontaneous process in a system at constant temperature and pressure?

I. $\Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$ II. $\Delta G_{\text{sys}} < 0$

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

66. What is ΔG_f° of HOCl(g) at 298 K?



$$K_{\text{eq}}(298 \text{ K}) = 0.089$$

- (A) -6.0 kJ mol⁻¹
(B) -3.0 kJ mol⁻¹
(C) 3.0 kJ mol⁻¹
(D) It cannot be determined from the information given.

67. The standard enthalpy of formation, ΔH_f° , for HCOOH(l) is equal to the standard enthalpy change for which reaction?

- (A) $\text{C}(\text{g}) + 2 \text{H}(\text{g}) + 2 \text{O}(\text{g}) \rightarrow \text{HCOOH}(\text{l})$
(B) $\text{C}(\text{s}) + \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{HCOOH}(\text{l})$
(C) $\text{C}(\text{g}) + \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{HCOOH}(\text{l})$
(D) $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{HCOOH}(\text{l})$

68. 40.0 mL of 0.200 M aqueous NaOH is added to 200.0 mL of 0.100 M aqueous NaHCO₃ in a flask maintained at 25 °C. Neglecting the effects of dilution, what is q for this reaction?

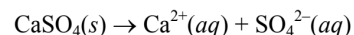
	ΔH_f° , kJ mol ⁻¹
OH ⁻ (aq)	-230
HCO ₃ ⁻ (aq)	-692
CO ₃ ²⁻ (aq)	-677
H ₂ O(l)	-286

- (A) -41 J (B) -74 J (C) -330 J (D) -820 J

69. In a well-insulated vessel, 50.0 g ice at 0.0 °C is added to 350. g water at 32.0 °C. What is the final temperature when the mixture reaches equilibrium? (The heat of fusion of ice is 334 J g⁻¹.)

- (A) 18.0 °C (B) 20.6 °C
(C) 22.0 °C (D) 28.0 °C

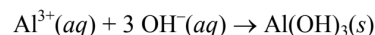
70. Which is the best explanation for the negative sign of ΔS° in the following reaction?



$$\Delta S^\circ = -143 \text{ J mol}^{-1} \text{ K}^{-1}$$

- (A) There are more ways of arranging the Ca²⁺ and SO₄²⁻ ions in aqueous solution than in the crystal lattice.
(B) Solid CaSO₄ is a network covalent solid, whereas it separates into ions in aqueous solution.
(C) Aqueous Ca²⁺ and SO₄²⁻ ions are tightly solvated, decreasing the number of ways of arranging water molecules when the solid dissolves.
(D) Calcium sulfate dissolves exothermically, leading to a net loss of entropy.

71. The K_{sp} of Al(OH)₃ is 2.0×10^{-31} at 298 K. What is ΔG° (at 298 K) for the precipitation of Al(OH)₃ according to the equation below?



- (A) -175 kJ mol⁻¹ (B) 14.7 kJ mol⁻¹
(C) 70.6 kJ mol⁻¹ (D) 175 kJ mol⁻¹

72. The bond dissociation enthalpies of the H-H bond and the H-Cl bond are 435 kJ mol⁻¹ and 431 kJ mol⁻¹, respectively. The ΔH_f° of HCl(g) is -92 kJ mol⁻¹. What is the bond dissociation enthalpy of the Cl-Cl bond?

- (A) 88 kJ mol⁻¹ (B) 96 kJ mol⁻¹
(C) 188 kJ mol⁻¹ (D) 243 kJ mol⁻¹